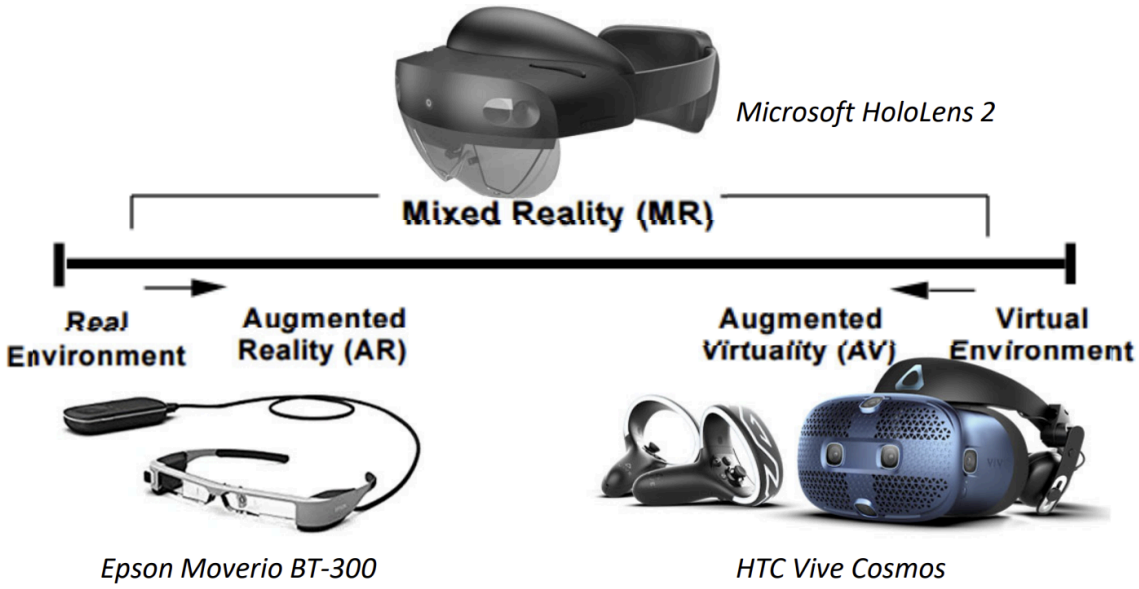
VR/XR For Manufacturing

MR, mixed reality, is a general term of any combination of the real world environment with virtual world graphics.



Virtual reality, VR is a pure virtual environment, with no real world environment elements.

Augmented reality, AR, superimposes virtual world graphics onto the real world environment. Typically, a camera is needed to record the real world environment, a processing unit, a display to superimpose the virtual elements, inputs, and feature/target markers, to figure out where to superimpose the virtual elements.

Typically, there are 3 common types of augmenting devices:

- Smartphones and Tablets
  - More common; more accessible for larger testing/deployment, generally higher computing power, larger screen provides easier inputs
  - However mostly non hands-free without a stand mount
- Smart Glasses & Wearable Headsets
  - hardware is specialised for complex AR applications, and allows hands free inputs
  - however has poorer battery life and is expensive and less accessible

- Spatial Displays
  - easily integratable into point-of-sales (locations where financial transactions happen), more flexible sensors and hardware, no smartphones required, higher quality graphics due to much more computing power
  - however expensive

As said earlier, AR needs to detect the real world environment to accurately/convincingly integrate graphics, which have multiple methods to do so:

- Marker Based
  - A predefined image is used as a marker to signify where the visuals should be overlayed
  - Is cheap, easy to implement and easy to interact with
- Marker-less
  - Based on Simultaneous Localization and Mapping (SLAM) technology, which allows graphics to move around independently without the need of markers
- Projection Based
  - directly overlays visuals in the real world via projectors, allowing for more seamless interactions
- Superimposition Based
  - provides alternate view of object by superimposing or replacing object with augmented view

Using AR technologies has many applications in different fields, such as

- entertainment
- architecture
- health & medicine
- aerospace & defense
- education
- data visualisation
- manufacturing

- some advantages of integrating AR into manufacturing include
  - Increasing manufacturing throughput
    - AR can provide clearer instructions for assembly and processes, which improves equipment uptime
  - accelerate training
    - AR can more clearly teach operators with immersive training
  - reduce costs
    - reduces mistakes in real world from clearer and immersive training
  - increase operator efficiency and safety
    - virtual safety warning may be added during operations and up to date instructions can be given with AR
- in manufacturing, applications include
  - showing complex assemblies
  - giving maintenance instructions
  - remote expert support/assistance
  - operator training
  - quality assurance
  - product design & development
  - automation

In the future, newer technologies like AI or IoT may be integrated alongside AR to further improve AR applications